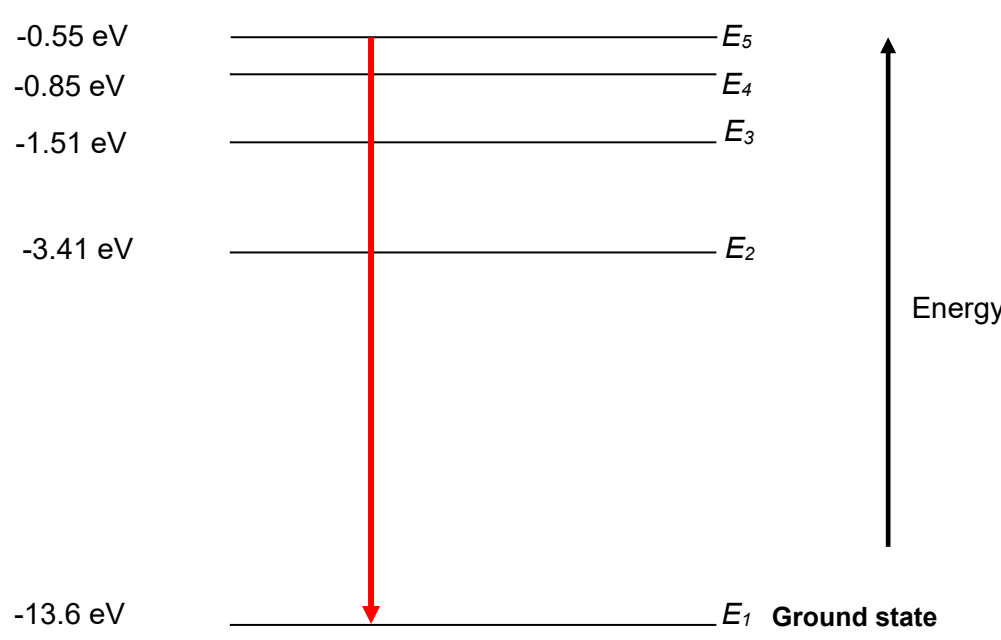
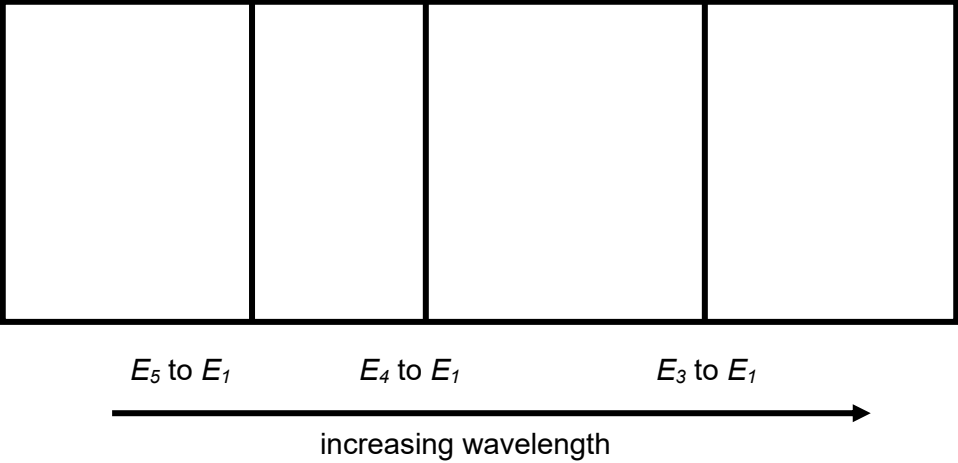


6a	Electrons from heated cathode bombarded electrons in hydrogen atoms/molecules. Causing electrons in hydrogen atoms/molecules to be excited from lower energy levels to higher energy levels. The excited electrons at higher energy levels subsequently de-excite and gives off energy in the form of photons, with energy exactly equal to the difference in energies between the energy levels. (The emitted photons thus made up the visible light).
6b	(Well-defined, distinct) Coloured lines on dark background.
6ci	 The diagram shows five horizontal lines representing energy levels, labeled E_1 through E_5 from bottom to top. To the left of the lines are their corresponding energy values in eV: E_1 is -13.6 eV, E_2 is -3.41 eV, E_3 is -1.51 eV, E_4 is -0.85 eV, and E_5 is -0.55 eV. A thick red arrow points vertically downwards from the E_5 level to the E_1 level. To the right of the lines, a vertical black arrow points upwards, labeled 'Energy'. The E_1 level is also labeled 'Ground state'.
6cii1	Energy of photon = Difference in energies between energy levels $\frac{hc}{\lambda} = [-0.55 - (-13.6)](1.6 \times 10^{-19})$ $\lambda = 95.3 \times 10^{-9} \text{ m}$
6cii2	10

6cii3	 Note: Labelling not required. The 2 shorter wavelengths should be closer. Shorter wavelengths to be located at the left.
6di	Energy of most energetic photon = biggest difference in energies between energy levels $\frac{hc}{\lambda} = [-0.55 - (-13.6)](1.60 \times 10^{-19})$ $\frac{hc}{\lambda} = 2.088 \times 10^{-18} \text{ J}$ Energy of photon = $\phi + KE_{\text{max}}$ Loss in KE = Gain in EPE $KE_{\text{max}} = eV_s$ Energy of photon = $\phi + eV_s$ $2.088 \times 10^{-18} = \phi + 2.45(1.60 \times 10^{-19})$ $\phi = 1.696 \times 10^{-18} \text{ J}$
6dii 1	$I_{\text{sat}} = \frac{n_e e}{t}$ $\frac{n_e}{t} = \frac{3.0 \times 10^{-6}}{1.60 \times 10^{-19}} = 1.875 \times 10^{13}$ $\text{Quantum yield} = \frac{\frac{n_e}{t}}{\frac{n_p}{t}}$ $\frac{n_p}{t} = \frac{\frac{n_e}{t}}{\text{quantum yield}} = \frac{1.875 \times 10^{13}}{9.2 \times 10^{-7}}$ $= 2.038 \times 10^{19}$
6dii 2	Reflection [1] or Low rate of collision between electrons and photons due to large empty spaces. [1] or

	Absorption of photons by electron does not result in emission as the electrons deeper within the metal loses too much energy on way to surface. [1] or Some of the photons have energy less than the work function [1]
6diii 1	Intensity is reduced (by 50%) when unpolarised radiation is polarized, or the number of photons incident per second is thus also halved (and therefore, the maximum number of electrons emitted per unit time and) hence saturation current is also halved (to 1.5 μA)
6diii 2	The electrons will be emitted with higher maximum kinetic energy as a smaller part of energy received from photon is used to overcome work function energy. (Use eqn to explain, can accept) As a result, the potential necessary to stop the fastest electron is thus also increased. Or As there was a good range of photons incident on the metal plate, with the lower of work function energy, a bigger group of photons will have more energy than the work function energy and therefore be capable of releasing electrons. As a result, the saturation current/ max current increases.
6div	De Broglie equation suggested that the photons of the light will have particle like behavior with a finite <u>momentum</u> . Since the photons experience a <u>change in momentum</u> upon impact with the metal surface, the metal surface exerts a force on the photons. By Newton's 3 rd law, the photons also exert a <u>force</u> on the metal surface. As pressure is force per unit area and force is rate of change of momentum, the impact of photons on metal surface will exert a force and hence pressure on the metal.